

Jan Ciara

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EDUCATION

Polish-Japanese Academy of Information Technology

Computer Science (Bachelor's Degree), Grade Average: 4.75/5.0

Warsaw

Expected 2028

EXPERIENCE

Java Developer Intern

Brown Brothers Harriman (BBH) – AI Lab

Mar 2026 – Present

Kraków, Poland

- Built backend features for a generative-AI document-processing platform (**Java / Spring**), routing uploaded files to stage-specific storage across the service and database persistence layers.
- Extended the **Java / Spring Boot** backend of an internal engineering skills-tracking tool, reworking a REST endpoint and its **JPA** associations to accept multiple entity links per request and to fix duplicate handling that was dropping existing rows.
- Developed and maintained a shared Jenkins CI/CD library in Groovy used across Java and Python microservices; consolidated **28** per-application deployment scripts into a single config-driven job promoting QC-validated images to production via Skopeo and OpenShift.
- Delivered end-to-end inference logging for two production ML models (**Python**), streaming raw inputs, model inputs, and predictions through **Kafka** into an **Oracle** monitoring store; resolved a record-alignment defect that was corrupting the logged data.
- Identified a flaw in the shared pipeline that masked failing test runs, allowing services with broken test suites to build and deploy to production; reworked the test stage to fail the build outright and traced the same issue in the Java pipeline.

Helpdesk Intern

Techtronic Industries (TTI)

Aug 2025 – Oct 2025

Warsaw, Poland

- Configured and deployed over **800** devices for a company-wide hardware refresh using Microsoft Intune and ManageEngine, alongside remote troubleshooting of user-reported incidents.

PROJECTS

LSM-Tree Storage Engine | Java 21, Gradle, JUnit 5, JMH | [GitHub](#)

- Built a persistent key-value store from scratch on an LSM-tree architecture (write-ahead log → memtable → immutable SSTables → streaming k-way merge compaction) with **zero database libraries**, covered by **104** tests.
- Designed a custom binary SSTable format with varint (LEB128) encoding, block indexing, and a hand-written Bloom filter at **10 bits/key**, reducing a point read from a full-file scan to a single **4 KiB** block and a read miss from **~100 µs to 0.5 µs**.
- Engineered crash safety without a MANIFEST file: atomic file replacement (tmp + fsync + rename) and a deletion ordering that guarantees an interrupted compaction cannot resurrect a deleted key, with the invariant locked down by a regression test.
- Profiled the read path with **JMH** and found that opening the file on every query accounted for **99%** of read time; switching to a persistent channel with positional reads yielded a **13x** speedup (**85 µs → 6 µs**).

Chess Engine | C++, CMake | [GitHub](#)

- Built a UCI-compliant chess engine reaching **2+ million** nodes/second, using bitboards and CPU intrinsics; move generation verified by **perft** against reference values up to depth 6.
- Implemented magic bitboards for sliding-piece attack generation, enabling **O(1)** move lookup.
- Optimized board representation by relocating repetition history out of the hot **Board** struct, shrinking it from **4,256 to 152 bytes** and yielding a **~4–5x** search speedup.

CERTIFICATIONS

CS50 AI: Introduction to Artificial Intelligence | [Certificate](#)

Harvard University

SKILLS

Languages: Java, Python, C++, C, SQL, Groovy, Bash

Backend & Testing: Spring Boot, Spring Data JPA, Hibernate, REST APIs, JUnit 5, Mockito, JMH

Data & Infrastructure: Oracle, Kafka, Git, Linux, Docker, Jenkins, OpenShift, Maven, Gradle, CMake, CI/CD, Nexus